Computers and Electronics in Agriculture 243 (2026) 111412

Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag

Original papers

Optimized aquaculture feeding through matched-filter audio signal processing and machine learning

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ARTICLE INFO

Keywords:
Audio signal processing
Aquaculture monitoring
Machine learning
Passive acoustic monitoring
Fish feeding intensity

ABSTRACT

Accurate feed management in marine aquaculture is critical for maximizing fish growth and minimizing environmental impacts. While previous approaches have leveraged acoustic monitoring and neural network-based classification to assess feeding, these methods often lack scalability and precision for industrial deployment. This study introduces a matched-filter audio signal processing technique, informed by domain knowledge, to continuously quantify fish feeding intensity in gilthead seabream (Sparus aurata) aquaculture. By extracting a species-specific bite acoustic template and applying matched filtering with sliding window aggregation, we generate a robust, continuous feeding intensity label from passive acoustic recordings. Machine learning regression models (XGBoost and Random Forest) validate the approach, demonstrating that environmental and biological variables explain 89% of the variation in feeding intensity. The proposed methodology offers a simple, scalable, and cost-effective solution for real-time feed optimization and welfare monitoring in aquaculture systems. By reducing data dimensionality and enhancing sensitivity to subtle behavioral changes, this framework supports the deployment of advanced data-driven monitoring tools and paves the way for practical integration in commercial aquaculture operations.

1. Introduction

Aquaculture is expanding rapidly, with mariculture emerging as a key sector in response to the rising global demand for sustainable seafood (FAO, 2024). This expansion presents new challenges and opportunities to improve fish growth and minimize environmental impact, particularly in high-value species such as gilthead seabream (Mhalhel et al., 2023). Accurate monitoring of feeding is a key factor in achieving these two goals (Price et al., 2015). Excess feed enriches the water with nutrients that accelerate eutrophication, algal blooms, and gas emissions, while oxygen is depleted and driving the system toward anoxic conditions (Fadum et al., 2024). Further, studies in seabream and seabass have indicated that a large share of the feed is left uneaten (Ballester-Moltó et al., 2017), highlighting the enduring challenge of overfeeding. Thus, continuous quantification of feeding intensity tracks real-time appetite, aligning feed delivery with actual demand (Zhao et al., 2024) rather than biomass-based tables.

Disease events have also become a central threat, carrying heavy economic and welfare costs (Naylor et al., 2021). Feeding intensity reflects metabolic condition and stress (Flores-García et al., 2022). Hence, continuous feeding intensity monitoring may serve as a sensitive indicator that can reveal shifts in appetite as an early welfare signal (Roberts, 2012; Aljehani et al., 2025).

Acoustic signals are promising tools for continuous quantification of feeding intensity in aquaculture (Li et al., 2024). Passive Acoustic Monitoring (PAM) methods capture characteristic sound patterns, providing a non-invasive approach to assess feeding behavior (Li et al., 2020). Recent studies have used PAM to *classify* fish feeding intensity into discrete categories. Zeng et al. (2023) applied acoustic signals with an audio spectrum Swin Transformer to classify feeding intensity into four levels: strong, medium, weak, and none. Du et al. (2023) used Mel spectrograms and a lightweight network to group feeding sounds into three classes: strong, medium, and none. Ma et al. (2024) used six-axis inertial sensor data and also proposed a method for classifying feeding intensity across similar categorical levels.

Within this classification paradigm for quantification of feeding intensity, Convolutional Neural Networks (CNNs) have been widely applied. Cui et al. (2022) proposed CNN6, which takes Mel spectrograms as input and achieved a mean average precision (mAP) of 0.74 on the

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https://doi.org/10.1016/j.compag.2026.111412
Received 28 October 2025; Received in revised form 5 December 2025; Accepted 3 January 2026
Available online 8 January 2026
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AFFIA3K dataset, a collection of 3000 labeled audio clips covering fish feeding intensities of none, weak, medium, and strong. Similarly, Li et al. (2025) reported that integrating acoustic spectrograms and pre-trained MobileNetV2 features with video inputs yielded a cross-modal framework achieving 95.4% accuracy, outperforming the audio-only (86.0%) and video-only (89.0%) models. These studies illustrate the promise of CNN-based approaches while also noting their reliance on high-dimensional inputs and considerable computational resources.

In practical aquaculture scenarios, where continuous monitoring must account for both cost and efficiency constraints, integrating a domain-aware data reduction scheme can introduce an additional layer of robustness to deep learning models. As was shown by Feldman et al. (2025) compressing the raw input signals into behaviorally meaningful representations, reduces the computational burden while preserving the information most relevant to feeding detection. Consequently, CNNs can achieve greater practicality and adaptability when complemented by dimensionality reduction mechanisms. The theoretical rationale for this claim lies in the CNNs' operation — their initial stage involves feature crafting, in which raw signals are processed and the network learns to extract the most informative features for classification or regression tasks. By enhancing the representation of the target phenomenon, in this case, feeding sounds, the learning process becomes simpler and more efficient, potentially leading to improved performance and several positive outcomes. Although *classification* models have demonstrated high accuracy in identifying feeding levels, their reliance on manual labeling may introduce biases and inconsistencies (Haliburton et al., 2025). More importantly, by constraining feeding behavior into fixed categories, they fail to reflect variance across different feeding intensities. The wide range of feeding intensities within a single class can lead to substantial errors in feeding-intensity estimation. In commercial settings, precise numerical quantification of feeding intensity is essential for feed optimization and early detection of welfare shifts that precede disease, especially given dense stocks, heavy daily feeding, and heightened susceptibility to rapid disease spread.

By contrast to the classification approach, a *regression* approach captures feeding intensities on a continuous scale, enabling more precise, data-driven decisions about feeding. Effective aquaculture management relies on continuously monitored biological and environmental processes, from biomass growth and cumulative feeding to water quality time-series (Rahmadani et al., 2024; Føre et al., 2016; Shreesha et al., 2023), making numerical regression signals a natural and necessary representation in the industry.

Furthermore, quantifying feeding intensity through regression holds strong potential as an early indicator of disease and optimized feeding appetite, as such patterns often evolve gradually rather than as sharp, discrete changes. Thus, these patterns are better captured by regression than by classification, highlighting the need for an efficient and automated approach.

One notable application of acoustic-based regression modeling in aquaculture is reported in studies of shrimp species such as *Litopenaeus vannamei*. Silva et al. (2019) examined feeding clicks in *Litopenaeus vannamei* and related them to mandible activity through regression, although the manual extraction of click counts limited the approach. Peixoto et al. (2020) extended this line of work by revealing an exponential relation between clicks per pulse train and signal duration. Together, these studies point to the value of regression for capturing feeding dynamics, yet the absence of automated methods for quantifying feeding intensity ultimately constrained both.

Using shrimps' feeding clicks is an example of prior biological knowledge, i.e., domain knowledge, where the clicks' sounds are a unique physiological feature. Using such knowledge allows a paradigm shift from Machine Learning (ML) to *machine education* (Kendler et al., 2022; Geltman et al., 2025). Machine education is an ML methodology that encompasses expert, problem-specific information beyond raw data (Barzamini et al., 2022). Applying machine education, rather than machine learning, often results in simpler ML architectures, smaller datasets that are required for training, and better generalization of the methodology across real-world conditions (Kendler et al., 2022; Geltman et al., 2025).

This study introduces a methodology for quantifying fish feeding intensity as a continuous numerical variable derived from PAM recordings analyzed with a *Matched-Filtering* (MF) approach informed by domain knowledge. MF is an optimized signal detection method for identifying known signal patterns in noise (Yaroslavsky, 2004), making it ideal for automated regression-based quantification of feeding intensity, as its underlying physics corresponds directly to the acoustic properties of fish feeding. Unlike simple correlation, which is restricted to local time-domain alignments and can fail when identical signals are phase-shifted, the matched filter operates in the frequency domain, where phase information is preserved, thereby eliminating the need for absolute timing. As a result, it will detect the presence of the reference pattern regardless of when it occurs in the signal, including situations when two or more patterns of interest overlap (Allen and Mills, 2004). This property is particularly relevant in aquaculture applications, where recordings are inherently noisy and sampling is finite. Yet, the fundamental principle of maximizing signal-to-noise ratio through coherent summation remains valid and ensures robust detection of feeding events.

Beyond its role in enhancing signal detectability, the methodology provides a considerable advantage through its natural capacity for data reduction. Raw acoustic recordings are sampled at tens of thousands of points per second, a resolution that is largely irrelevant to the temporal dynamics of fish feeding behavior. By applying matched filtering within a sliding window framework, this continuous high-frequency stream can be transformed into a single representative value corresponding to the duration of a feeding event. In this way, the approach demonstrates not only the effectiveness of the matched-filter in isolating a behavioral template but also achieves a substantial reduction of data, which can enable operation in real time and improve both the efficiency and the predictive performance of subsequent deep learning models. Moreover, rather than reducing events to coarse three or four level categories, this approach preserves within category variability by providing an interpretable numerical measure of feeding intensity, offering a more robust and accurate tool for feed optimization and welfare assessment (Pulido-Rodriguez et al., 2021).

Regression models built on key environmental and biological variables were then used to test the approach, showing that predicted feeding intensity aligned strongly with the underlying signals.

Building on this foundation, the acoustic monitoring framework developed here seeks to address the constraints and requirements that are characteristic of aquaculture systems. The approach is particularly tailored for aquaculture applications in high-density settings, such as Recirculating Aquaculture Systems (RAS) and open sea cages, where large populations must be monitored continuously, while enabling a non-intrusive, simple, low-cost acoustic scheme suited for routine operation.

2. Methodology

2.1. Systematic overview

Feeding behavior was monitored over 19-day period through continuous audio recordings of *gilthead seabream* maintained under fixed feeding schedules. The complete workflow is summarized in Fig. 1. The light-blue solid frame outlines the data-acquisition stage, in which hydrophones were positioned inside the tanks and continuous acoustic records were collected. An example of the raw waveform is shown to illustrate the character of the signal prior to any processing.

The orange dashed frame presents the audio-signal preprocessing stage. Here, the recordings undergo spectral-gating noise reduction followed by high-pass filtering to isolate the frequency content associated with feeding activity. The red dashed frame describes the

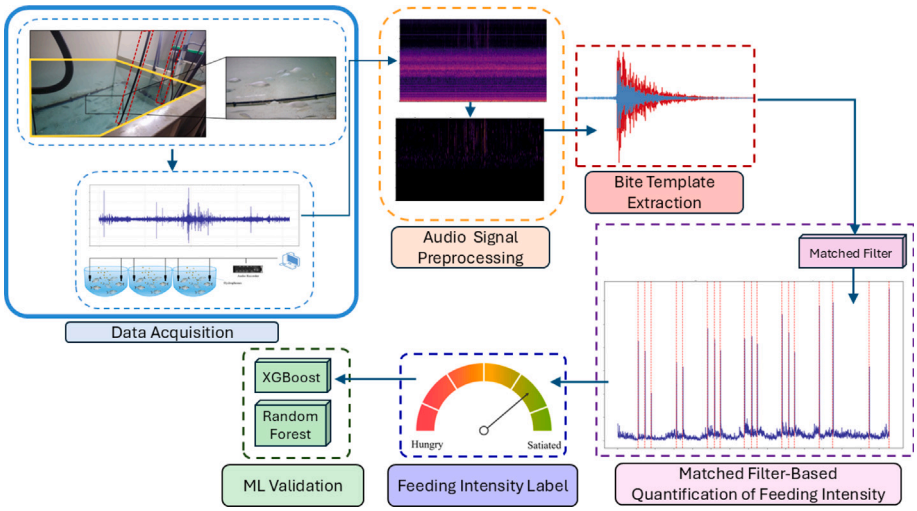

Fig. 1. Methodology workflow. Continuous 19 days audio recordings were collected from three fish tanks using hydrophones. Signals were preprocessed to reduce noise. A bite template was extracted and applied in a matched-filter analysis combined with a sliding window aggregation to detect and quantify feeding events intensities. Finally, the resulting feeding intensity label was validated against environmental and biological variables using XGBoost and Random Forest regression machine learning models.

bite-template extraction step, where a representative feeding click is manually identified and used as a biological template for subsequent matched-filter computation.

The matched-filter and sliding-window operations are shown in the purple dashed frame. The matched-filter, presented in the solid purple box, produces a continuous numerical trace that reflects the temporal density of bite-like events. This trace is then aggregated by a sliding window mechanism, generating the feeding intensity label depicted below the purple frame. The hungry-satiated gauge in the blue dashed frame represents the final numerical value of this label for each feeding event, providing an interpretable measure of feeding intensity. Finally, the green dashed frame outlines the machine learning validation stage, where Random Forest and XGBoost models are applied to evaluate how well environmental and biological variables explain the derived feeding intensity signal.

2.2. Data acquisition

The experimental system at the National Center for Mariculture in Eilat, Israel consisted of three identical flow-through tanks (2.0 × 1.0 × 1.0 m) with continuous aeration, each stocked with 30 gilthead seabream *(Sparus aurata)*, as shown in Fig. 2. This setup reproduced key acoustic characteristics of commercial RAS environments, including continuous water flow, aeration, operational equipment noise, and the absence of acoustic isolation.

At the start of the experiment, all fish were three months old (estimated average weight 40 g) and originated from the same hatching batch. Each tank was equipped with two hydrophones (AS-1 hydrophone, Aquarian Audio & Scientific, Anacortes, WA, USA) positioned on opposite sides to minimize spatial noise variation and enable independent validation of feeding sound detections. The hydrophones were connected to phantom-powered preamplifiers (PA6, Aquarian Audio & Scientific, Anacortes, WA, USA). Fish were fed one to three times daily at consistent times, receiving a total daily ratio equal to 2% of the total biomass using 3 mm commercial sinking pellets (Raanan Fish Feed Ltd., Israel). All feed was fully consumed. Artificial lighting simulated constant day-night cycles, with lights on at 06:00 and off at 17:00. Continuous acoustic signals were logged to memory cards using a Zoom F8n Pro recorder (Zoom North America, Hauppauge, NY, USA) at a sampling rate of 48 kHz and a bit depth of 24 bit.

In our experiment, as common in aquaculture systems, gilthead seabream (*Sparus aurata*) are typically fed industrially processed sinking pellets, which produce a distinct cracking sound during feeding. This consistent acoustic signature across individuals provides a reliable biological template, making it well-suited for matched filtering. Each feeding event generates a brief impulsive click with a consistent waveform and characteristic time–frequency energy distribution. Consequently, bite clicks that align with the template are strongly amplified, whereas broadband noise from pumps, aeration, and water circulation remains incoherent and is suppressed. This methodology is also applied for example in seismology, where matched-filtering is used to reliably detect and localize weak microseismic events within continuous noisy recordings (Eisner et al., 2008). In this study, the characteristic clicking sounds emitted by gilthead seabream during feeding were selected as the reference template, as their acoustic consistency provides a reliable basis for detection across individuals.

2.3. Acoustic signal preprocessing

The recorded acoustic signal contained background noise from operational and electronic equipment, as well as continuous water inflow and aeration. As gilthead seabream feeding sounds occupy a defined frequency band, targeted signal processing was needed to extract the relevant information. The signal was first transformed into the frequency domain using the Short-Time Fourier Transform (STFT), with a Hann window, a frame size of 2048 samples and 50% window overlap. Spectral gating was then applied to reduce noise across all frequencies, followed by a high-pass filter that removed low-frequency components and enhanced the higher frequency feeding signatures.

The STFT is a standard technique for converting time-domain signals into the frequency domain. It generates a spectrogram by partitioning the signal into overlapping windows and computing a Fourier Transform to each segment, thereby illustrating temporal changes in spectral energy. This representation enables the detection of short-duration acoustic events such as feeding events. In this study, STFT was implemented in Python using the *SciPy* library. This transformation is formally expressed in Eq. (1).

$$\hat{S}[f,t] = \mathrm{STFT}[f,t] = \sum_{n=0}^{L-1} s[n] \cdot w[n-t]\, e^{-j2\pi f n}, \tag{1}$$

where $s[n]$ is the input signal, $w[t]$ the analysis window centered at time t, f the frequency bin, n the time index within the window, and L is the window length. (The inverse transform, $iSTFT[f,t]$, is given in Appendix A)

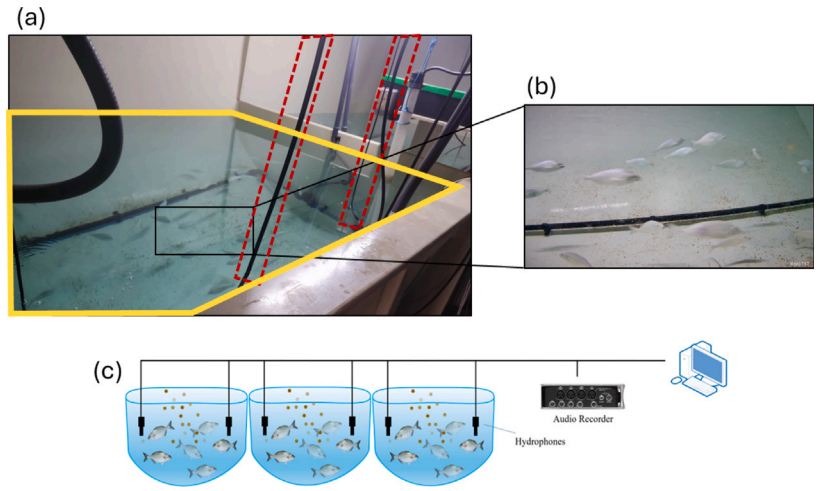

Fig. 2. Experimental setup. (a) Photograph showing one of the three tanks, with boundaries marked in yellow and hydrophones shown by red dashed lines. (b) Close-up view of fish activity inside the tank. (c) Schematic of the complete experimental system, including all three tanks, hydrophones, the audio recorder, and the data logging computer used for data collection.

Spectral gating is a noise reduction technique that reduces noise in each frequency band. The noise profile was estimated from a one-second reference extracted from each forty-minute segment. For each band, the amplitude standard deviation is computed, and components with magnitudes below 1.5 standard deviations are suppressed. This thresholding operation is formally defined in Eq. (2), where frequency bins falling below the threshold are set to zero, thereby enhancing acoustic signal clarity while minimizing noise.

$$\hat{X}(f,t) = \begin{cases} X(f,t), & \text{if } |X(f,t)| \geq T(f) \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

where $\hat{X}(f,t)$ is the gated spectrogram, $X(f,t)$ is the original spectrogram, and $T(f)$ is the threshold set to 1.5 standard deviations of the amplitude within each frequency band. This process was implemented in Python using the *noisereduce* library (version 3.0.3) (Sainburg et al., 2020).

Following spectral gating, residual low-frequency noise persisted, mostly originating from equipment and the surrounding tank environment. Because gilthead seabream feeding sounds occurred at higher frequencies, a 2048 Hz high-pass fifth order Butterworth filter with normalized cutoff 0.0853 and zero phase forward and backward implementation was applied to isolate the relevant signal and ensure effective removal of low frequency noise. The cutoff frequency was selected by jointly inspecting spectrograms and synchronized video recordings, which confirmed that frequencies below 2048 Hz did not contain feeding sounds.

2.4. Matched-filter design

Feeding intensity was quantified using a matched-filter algorithm, a signal processing method optimized for the detection of predefined patterns within noisy environments.

The matched-filter operates by transforming both the incoming signal and the predefined template into the frequency domain and aligning their spectral content through phase conjugation. In this operation, the phase of each template component is inverted, allowing frequency components in the signal that correspond to the template to become phase-aligned rather than dispersed. When alignment occurs, these components reinforce one another, generating a pronounced temporal peak that marks the presence of the target acoustic pattern. Non-matching background noise, which exhibits random and uncorrelated phase structure, fails to align and therefore contributes minimally to the output. Through this combination of spectral rephasing and sliding temporal correlation, the matched-filter selectively amplifies template-consistent structure while suppressing noise, yielding a robust detection mechanism for template identification within complex and noisy signals.

In this study, the reference pattern was defined as a single "click" bite sound produced by gilthead seabream during feeding. Subsequent detections were aggregated into feeding events using a sliding window procedure. The template, presented in the time domain in Fig. 3a, has a maximum power frequency of 4462 Hz and a spectral centroid of 5138 Hz. A detailed description of the bite template extraction procedure is provided in Appendix B.

Due to biological variation, templates may vary slightly among individuals of different weights and ages, as larger fish exert greater jaw force and produce stronger bite signals. Nevertheless, the biomechanical process underlying jaw movement is consistent across gilthead seabream individuals, ensuring that the fundamental principles of the matched-filter technique remain valid.

To further illustrate age-related differences, Fig. 3b presents a zoomed-in comparison between the bite waveform template derived from a young fish and a waveform extracted from an older individual. The x-axis represents the sample index at a 48 kHz sampling rate, and the y-axis indicates the signal amplitude. Both young and older fish show a comparable waveform decay, beginning approximately 4000 samples (≈80 ms) after the peak, with similar decay rates. These similarities demonstrate that the underlying acoustic characteristics remain consistent across age groups, supporting the use of a template obtained from any age group as a robust indicator of gilthead seabream feeding sounds.

Nevertheless, for optimal template matching across age groups, it is preferable to update the reference templates according to the specific cohort sampled.

After extracting the bite sound template, both the template and the recorded signal were transformed into the frequency domain using the STFT (Eq. (1)) with a window size of 1024 and a hop size of 512. Detailed parameter settings for the signal preprocessing and matched-filtering are summarized in Appendix A, Table A.1, which lists all parameters applied throughout the analysis. The template in the frequency domain is then conjugated and is multiplied element-wise with the signal's frequency response.

For $s(t)$ the input signal, $w(t)$ is the matched-filter template, and $STFT\{w[t]\}^*$ the complex conjugate of the template, the matched-filter output $M(t)$ is computed as shown in Eq. (3). The output then emphasizes segments of the signal that closely resemble the template.

$$M(t) = \|iSTFT\{STFT\{s[t]\} \cdot STFT\{w[t]\}^*\}\| \tag{3}$$

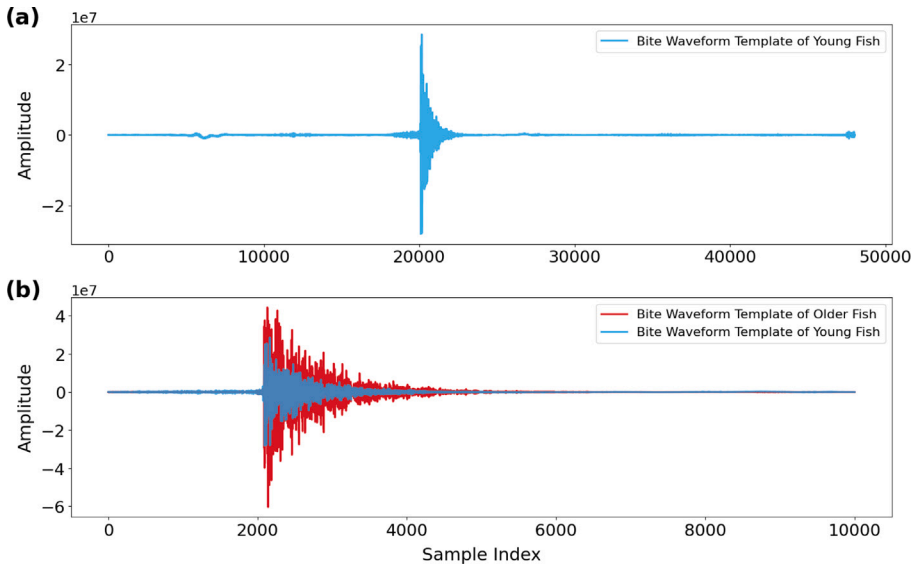

Fig. 3. Time-domain bite waveforms of gilthead seabream. (a) waveform of a single bite sound from a young fish used as the template in the matched-filter analysis. (b) a zoomed-in comparison of bite waveforms from young and older fish, illustrating the similarity in signal characteristics across ages.

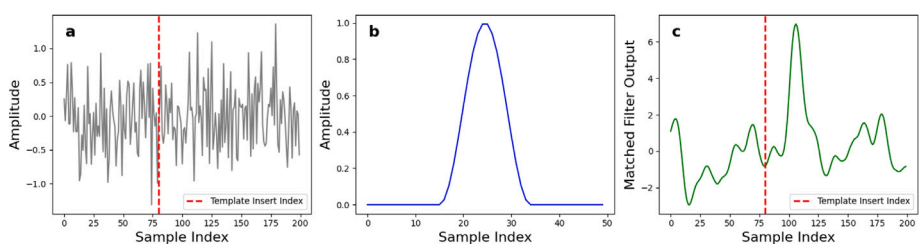

Fig. 4. Principle of matched-filtering. (a) Noisy signal with embedded pulse. (b) Predefined template. (c) matched-filter output showing a distinct correlation peak offset from the signal insertion point by the template duration.

The norm applied in Eq. (3) ensures that the matched-filter output reflects the signal's magnitude.

For the audio sensor-based system described in this study, matched filtering operates directly on the one-dimensional waveform using lightweight FFT convolution, yielding a compact, physically informed representation that is more efficient than extracting features from high-dimensional spectrograms with deep CNNs. CNN approaches remain valuable for learning complex patterns, with the matched filter providing an efficient front end that reduces input complexity and facilitates their deployment when needed.

Fig. 4 illustrates the MF process through three subplots, where the x-axis in all panels represents the sample index. The left panel shows a noisy signal into which a pulse-shaped template was embedded at sample index 80, marked by a red dashed vertical line. The middle panel displays the predefined template. The right panel presents the matched-filter output, where the y-axis represents the filter response magnitude.

In the noisy signal, the embedded template is visually indistinguishable because of the high noise level. However, the matched-filter output reveals a distinct peak at sample index 110, corresponding to the insertion point (index 80) offset by the template duration. The peak amplitude reaches approximately 6, standing out clearly from the background signal, which remains below 2. This illustrates the matched-filter's ability to detect predefined patterns embedded in noise by amplifying their correlation response at the correct temporal location.

In aquaculture environments, during vigorous feeding events, multiple fish may bite simultaneously, producing overlapping acoustic events. According to Parseval's theorem, total signal energy is preserved between the time and frequency domains. Therefore, overlapping MF responses do not eliminate information but instead increase the energy within the relevant frequency bands, thereby maintaining accurate bite count estimation in the frequency domain. Moreover, since the method ultimately counts signals above the threshold, in cases of perfect overlap, even if the energy is doubled, it will still be counted as a single event. Hence, the matched-filter remains effective under overlapping because it emphasizes spectral energy alignment rather than strict phase coherence, enabling detection of characteristic bite signals. Its performance does degrade under severe overlap, but within practical limits it maintains detectability. Appendix C presents a quantitative assessment of its sensitivity to overlapping bite signals.

Sliding window analysis is a standard approach for processing time series data, in which a fixed length window slides across the signal with a defined step size. In this study, the method was applied to the matched-filter output. At each step, the algorithm counts the number of values within the window that exceed a predefined threshold, producing aggregated values that reflect the intensity of feeding events, as defined in Eq. (4).

$$I(t) = \sum_{i=0}^{W-1} [M(t+i) > \theta] \tag{4}$$

where $I(t)$ is the feeding intensity score at time t, $M(t+i)$ is the matched-filter output at time $t+i$, W is the window length, and θ is the detection threshold. The expression $[M(t+i) > \theta]$ equals 1 when the condition is met and 0 otherwise, effectively counting matched-filter values that exceed the threshold within the window.

To enable continuous analysis of the 19 day dataset, the MF output was averaged over non-overlapping windows of 1000 samples,

reducing the temporal resolution to 48 samples per second. This downsampling step reduced temporal redundancy, preserving acoustic information while aligning the signal representation with the biologically relevant timescale of fish behavior. Each daily time series feeding intensity label, defined as the sum of MF outputs exceeding the threshold within each window, was normalized to the day's maximum value, producing a relative measure of feeding intensity.

2.5. Machine learning performance evaluation

Following the described workflow, each recorded feeding event was assigned a continuous numerical label representing its feeding intensity. For each feeding event, three environmental and biological numeric variables were considered: (i) fish age (days since the start of the experiment), (ii) feeding time during the day, expressed in 24-hours format with decimal fraction denoting minutes (e.g., 14.3 corresponds to 14:18), and (iii) time since previous feeding (minutes). The feeding intensity label of each event was defined as the peak value of the summed MF outputs exceeding the detection threshold within the corresponding feeding episode. This peak value is the feeding intensity label and was used as the target feature for the regression model.

Correlations between the numerical variables, including normalized feeding intensity label, time from feeding (min), and fish age (days) and time of day (hours) were analyzed using the Pearson correlation coefficient.

To evaluate the alignment of the feeding intensity label with environmental and biological variables, regression XGBoost and Random Forest models were applied. Tree based regression is especially useful here, since its hierarchical partition of the feature space yields explicit decision paths that support transparent physical interpretation. By tracking impurity reduction at each split and aggregating it into feature importance scores, the models provide a clear, physically grounded explanation of the relative contributions of individual environmental and biological variables to feeding intensity predictions. The dataset was randomly divided into training (80%) and testing (20%) subsets. Models hyperparameters were optimized using a grid search with 10-fold, KFold cross-validation, applying Mean Squared Error (MSE) as the evaluation metric. The parameter grids used for model optimization are described in Appendix A, Table A.2.

All models were implemented in Python using the *scikit-learn* library. The optimized XGBoost model employed a maximum depth of the individual regression estimators, max_depth = 5; the fraction of samples to be used for fitting the individual base learners, subsample = 0.8; the decrease in the contribution of each tree as the algorithm progresses, learning_rate = 0.1; and the number of boosting stages to perform, n_estimators = 50. The optimized Random Forest model used max_depth = None; the minimum number of samples required to split an internal node, min_samples_split = 2; and the number of trees in the forest, n_estimators = 50.

3. Results

3.1. Acoustic signal preprocessing

The preprocessing stage effectively enhanced the clarity of the acoustic signal by reducing broadband and low-frequency noise. The outcome of this process is illustrated in Fig. 5, which presents the Mel spectrogram of a representative feeding event before and after noise reduction. The spectrogram provides a compact time–frequency representation of the signal, generated using the Python library *Librosa*. The x-axis denotes time in minutes over a 40-minute recording, while the y-axis represents Mel frequencies ranging from 0 to 17 kHz, distributed on the Mel scale to provide finer resolution at lower frequencies and compressed resolution at higher ones. The color bar indicates amplitude in decibels (dB), where lighter shades correspond to stronger amplitudes (0 dB) and darker shades to weaker amplitudes (–80 dB).

Before preprocessing, substantial spectral energy appeared between 0–4000 Hz, with amplitudes of 0 to –50 dB persisting throughout the recording. After applying spectral gating and high-pass filtering, this energy was largely attenuated to around –80 dB, while the relevant high-frequency bite signals remained distinctly visible. For example, strong high-frequency signals between 24 and 26 min, previously masked by noise, became clearly recognizable following processing. These results confirm that the adopted combination of spectral gating and high-pass filtering effectively suppresses background noise while preserving the frequency bands associated with feeding sounds.

3.2. Matched-filter output

The matched-filtering results are presented in Fig. 6, which compares the raw preprocessed waveform (Fig. 6a) with the MF response to bite sounds (Fig. 6b) over a 7-days period, while Fig. 6c shows a zoomed-in view of the matched-filter output over a 40-minute segment, highlighting the 0.01 detection threshold. In both plots a and b, the x-axis represents time in days, while the y-axis of the raw waveform indicates amplitude, whereas the MF response ranges from 0 to 1.1, with higher values corresponding to stronger similarity to the bite template. The raw signal was influenced by transient environmental noise, leading to high-amplitude fluctuations unrelated to feeding. For example, at day 2.5 a spike of $\pm 1.4 \times 10^6$, probably caused by machinery, was present in the raw waveform but absent in the matched-filter output. Similarly, at 0.75 days the raw waveform exhibits a spike of $\pm 0.3 \times 10^6$, whereas the corresponding MF value (≈ 0.4) remains lower than that of true feeding events. These comparisons demonstrate that the broad amplitude range of the raw waveform masks subtle acoustic signatures of feeding, whereas the MF output projects the signal onto a domain specific scale that suppresses irrelevant fluctuations and amplifies the distinctive features of feeding behavior. Consequently, the MF response provides a clearer and more interpretable representation of feeding behavior.

The specificity of the matched-filter template was assessed by comparing its performance to 20 randomly selected one second templates extracted from random periods throughout the day. For each template, the MF output was used to calculate the signal-to-noise ratio (SNR), defined in Eq. (5).

$$\mathrm{SNR} = \left(\frac{\mu_{\text{sig-top}}}{\mu_{\text{noise}}} \right) \cdot \sigma_{\text{sig}} \tag{5}$$

where $\mu_{\text{sig-top}}$ is the mean of the top k values in the signal window (with k = 48,000, corresponding to one second at a 48 kHz sampling rate), μ_{noise} is the mean of the noise window, and σ_{sig} is the standard deviation of the signal window. The selected bite template achieved an SNR up to 16 times higher than that of the random templates, confirming its strong specificity. This high specificity indicates that the template accurately captures the acoustic signature of individual bites.

Because each tank contained only about 30 fish, the low stocking density made such overlap unlikely, especially given that at 48 kHz most bite energy was confined to a narrow millisecond window. However, in less favorable conditions such overlap may occur, leading to signal saturation during high-intensity events. As the aim of the feeding analysis is to label the overall feeding intensity of each event rather than individual bites, this limitation is of minor practical significance.

The hyperparameters including window length, step size, and detection threshold were optimized using a grid search approach in which the receiver operating characteristic (ROC) curve and its corresponding area under the curve (AUC) served as the performance metrics. The ROC framework is particularly suited for this task, as it provides a principled means of evaluating classifier performance by quantifying the trade-off between true-positive and false-positive rates.

This approach is especially relevant in the present study, where the objective is to reliably detect genuine feeding events while suppressing spurious detections originating from background noise. As emphasized

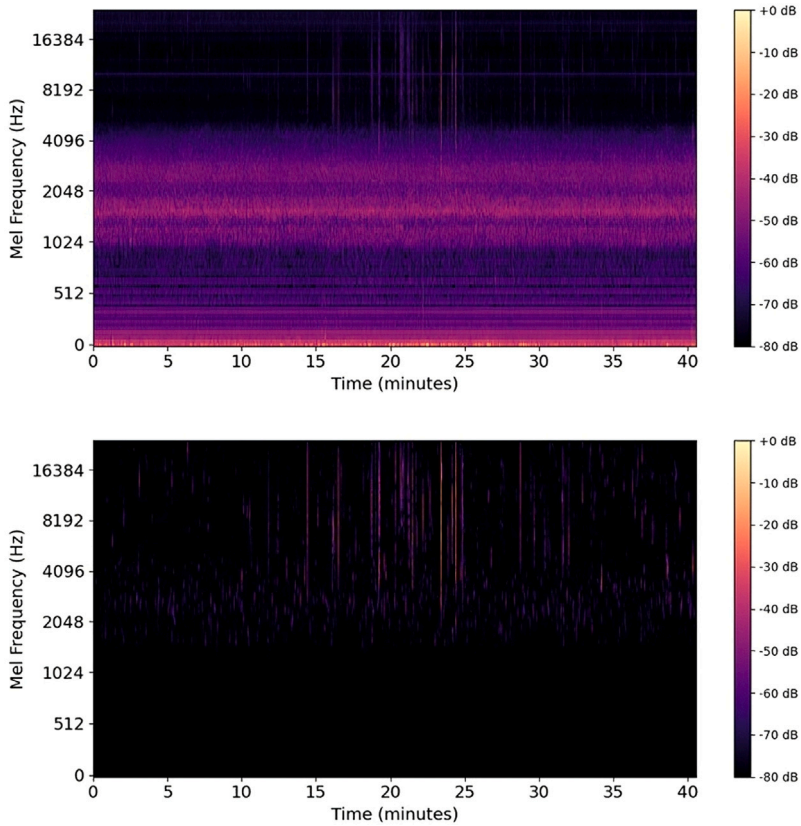

Fig. 5. Mel spectrograms of feeding event before (top) and after (bottom) noise reduction. Spectral gating and high-pass filtering enhance signal clarity by reducing background noise.

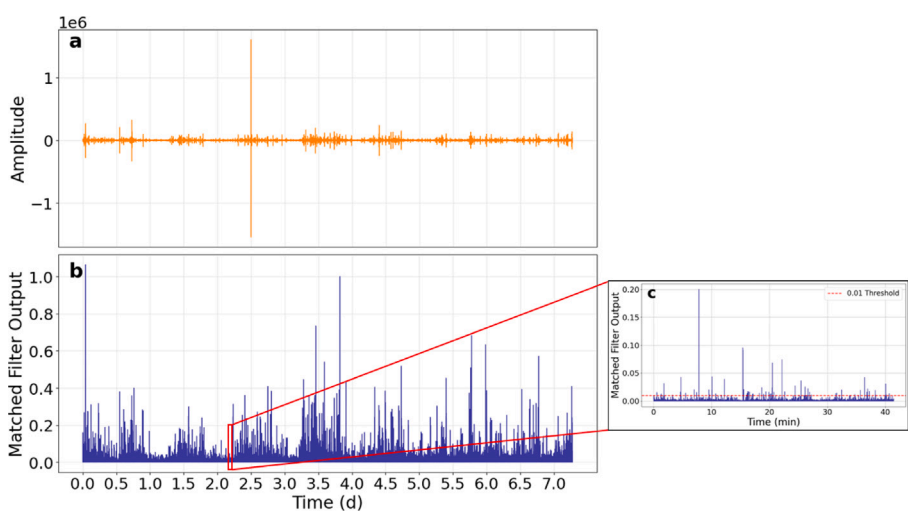

Fig. 6. Comparison of signals before and after matched-filtering. (a) Raw waveform after preprocessing without matched-filtering. (b) matched-filter response to the bite sound template over a 7 days period, illustrating the enhanced detectability of feeding events. (c) Zoomed-in view of the matched-filter output over a 40-minute segment showing the 0.01 detection threshold.

by Fawcett (2006), ROC analysis offers both an intuitive visualization and a statistically grounded measure of discriminative ability, ensuring that the selected hyperparameters are optimized for extracting meaningful feeding activity from acoustic signals.

ROC curves were generated by defining ground-truth feeding events based on the experimental log, where MF detections with SNR above 10 were treated as true positives. Noise was estimated from the mean signal amplitude measured outside feeding periods.

Step sizes from 1 to 60 s were evaluated in 5 s increments, revealing negligible AUC variation (less than 0.01) when window length and threshold were held constant. A six seconds step size was therefore selected, since feeding events evolve over minutes, making adjacent short steps yield nearly identical MF responses and little change in performance. Following step size selection, further optimization examined the effects of window length and MF threshold on classification performance.

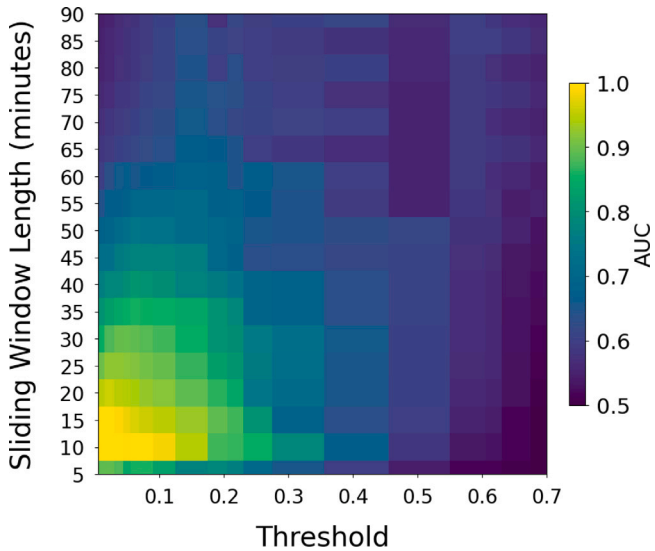

Fig. 7. Grid search optimization results for sliding window hyperparameters. The heatmap shows AUC values obtained from ROC analysis across combinations of window length (minutes) and MF threshold. Optimal performance was achieved with shorter window lengths and lower threshold values.

Fig. 7 presents the grid search optimization results for the sliding window parameters, evaluated using ROC–AUC performance. The x-axis indicates the MF threshold, the y-axis represents the sliding window length (minutes), and the color bar shows the corresponding AUC values.

The highest AUC values were observed for window lengths ranging from 8 and 17 min and MF thresholds between 0.01 and 0.1. Within this parameter range, classification performance reached perfect with no false positives. Increasing either the window length or the threshold beyond this range led to a gradual decline in AUC performance. For example, window lengths of 17–35 min combined with thresholds of 0.15–0.3 produced AUC values between 0.7 and 0.9. When window lengths exceeded 35 min and thresholds rose above 0.3, AUC values decreased to 0.5–0.7, indicating information loss and reduced sensitivity to feeding events.

An exception occurred for window lengths shorter than 7 min, where AUC declined from 1.0 to approximately 0.85. This finding reflects the biological characteristics of seabream feeding behavior, which typically lasts longer than five minutes. Consequently, shorter windows failed to capture the full duration of feeding events, resulting in reduced detection accuracy.

These results confirm that the selected parameters are consistent with the temporal dynamics of seabream feeding events and support the accurate estimation of the feeding intensity label. They further demonstrate the importance of tailoring model parameters to the biological characteristics of the studied system.

Following the discussion above, for the sliding window processing, a 10 min sliding window with a 6 s step size was used, with a threshold of 0.01, as shown in Table A.1.

3.3. Quantification of feeding intensity

Using these optimized parameters, the sliding window output was aggregated to derive a continuous feeding intensity label. Fig. 8 illustrates seven days aggregation results. The x-axis represents time (days), while the y-axis indicates the feeding intensity label, computed as the number of samples exceeding the MF threshold per window. Red dotted vertical lines mark logged feeding events, typically one to three times per day.

As shown, the feeding intensity labels corresponding to feeding events were markedly higher than background activity. For example, between days 0.5 and 1, three peaks in the feeding intensity label are observed, aligning with the logged events: morning (420), noon (390), and evening (200), compared to background levels averaging around 20. A similar pattern appears between days 4.5 to 5, with morning, noon, and evening labels of 550, 480, and 390.

Throughout the observation period shown in Fig. 8, non feeding activity yielded label values between 0 and 90, whereas feeding events ranged from 200 to 650, indicating clear separation. The results also capture diurnal feeding dynamics, with morning feedings typically more intense than those later in the day, consistently observed across days 0–5.

Moreover, an upward trend in feeding intensity was observed over time, consistent with the juvenile growth stage. Label values rose from 200–420 on days 0–2 to over 570 by day 5, reflecting increased feeding activity with rapid growth. These observations demonstrate that the proposed methodology provides a reliable means of detecting and quantifying feeding intensity, capturing both short-term behavioral patterns and long-term biological trends.

To complement the 7 days example overview, a closer examination of a single day provides a detailed view of the method's temporal resolution, as shown in Fig. 9. The x-axis represents the time of day, showing three feeding events at 08:00, 12:30 and 16:30. The y-axis indicates the feeding intensity label, while red dashed vertical lines mark the logged feeding times. A closer view of each feeding peak reveals a sharp rise and rapid decay, reflecting the short duration of each feeding event. The peaks in the plot appear slightly before the logged feeding times because the sliding window aggregates the signal over the preceding 10 min interval, which approximately corresponds to the feeding period. As in the multi-day example shown in Fig. 8, the feeding peaks are substantially higher than non feeding activity, reaching values of 400–500 in the feeding intensity label compared with less than 90 during inactive periods.

The resulting continuous and quantitative feeding intensity label demonstrates the effectiveness of the proposed methodology in accurately quantifying feeding behavior and distinguishing it from background activity. Its reliability was further evaluated by validating the feeding intensity label against environmental and biological variables using statistical analysis and ML models.

3.4. Machine learning performance evaluation

The feeding intensity label was validated against environmental and biological variables to ensure its robustness, reliability, and correspondence with real environmental conditions. Fig. A.1 in Appendix A presents the distributions of the feeding intensity label together with three explanatory variables, providing an overview of the experimental conditions under which feeding behavior was observed. A statistical examination of the feeding intensity label, illustrated in Fig. 10, reveals how feeding behavior varies across different times of day. The x-axis represents the time of day, categorized as morning (06:00–10:00), noon (10:00–14:00), and evening (14:00–18:00), while the y-axis shows the corresponding normalized feeding intensity values. Vertical lines denote the standard deviation of each category, illustrating variability in feeding intensity throughout the day. Morning feedings displayed the highest normalized intensity (mean = 1.00 ± 0.01), followed by noon (mean = 0.68 ± 0.20) and evening (mean = 0.62 ± 0.20). These values indicate a consistent and statistically significant difference between morning feedings and those occurring later in the day. This pattern aligns with biological expectations, as morning feedings occur after the overnight fasting period, resulting in elevated appetite and feeding activity. Such correspondence confirms that the proposed methodology reliably captures temporal variations in feeding behavior and is well suited for analyzing fish behavioral dynamics.

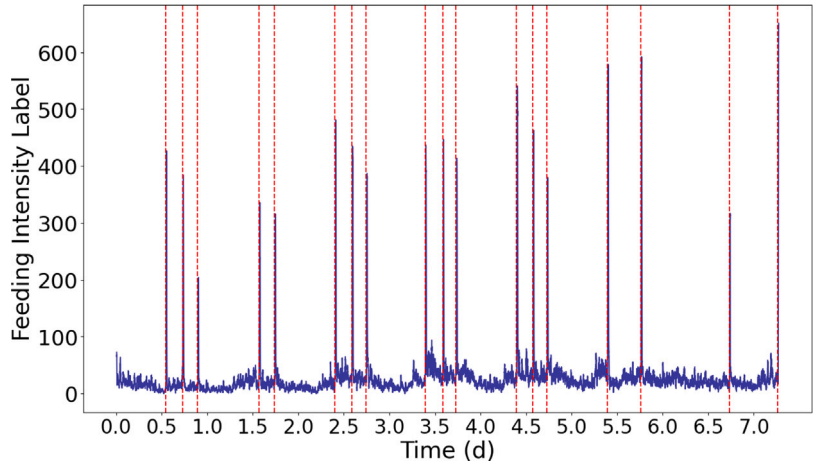

Fig. 8. 7 days example of the feeding intensity label generated using the sliding window applied to the MF response. Red dashed lines indicate feeding events recorded during the seven days period shown. Distinct peaks corresponding to feeding events are clearly visible within this period.

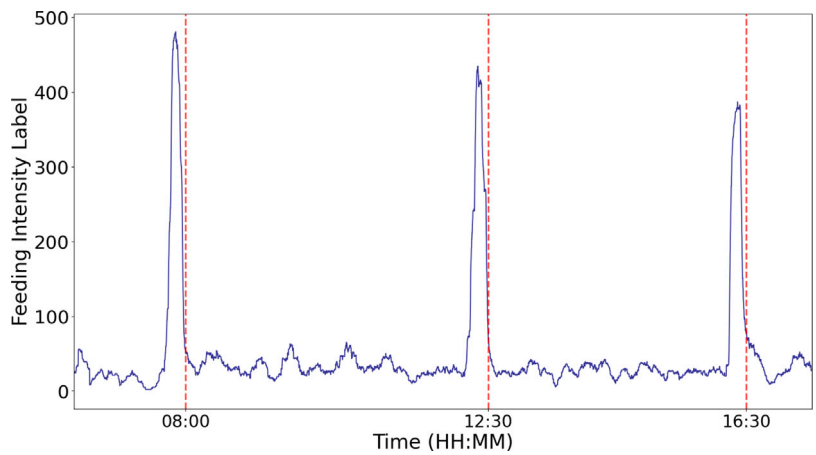

Fig. 9. Feeding intensity label over a single day. The blue line represents the number of samples exceeding the sliding window threshold, while the red dashed lines indicate the scheduled feeding times (08:00, 12:30, and 16:30). Distinct peaks align with the feeding periods, demonstrating the effectiveness of the methodology in capturing feeding activity.

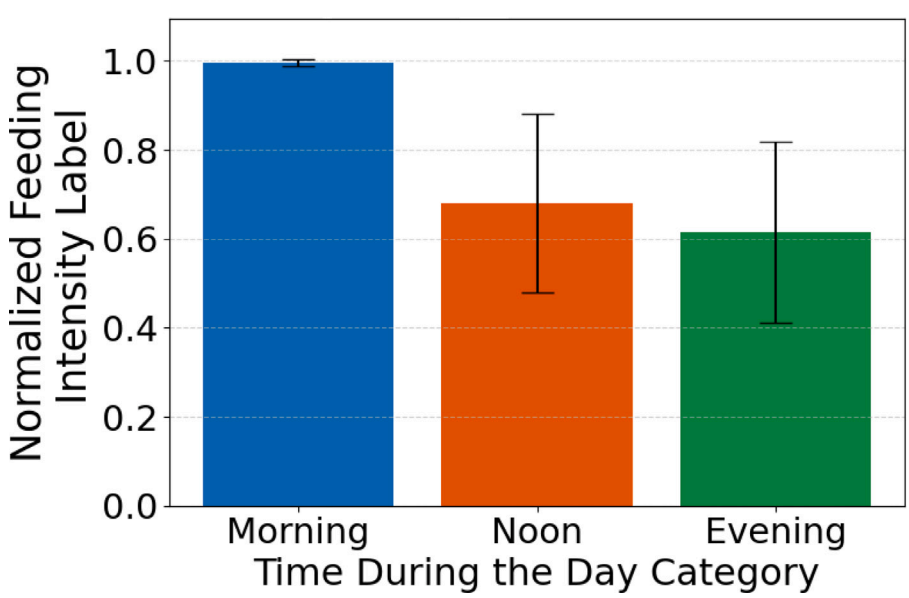

Fig. 10. Normalized feeding intensity labels across different times of day: morning (06:00–10:00), noon (10:00–14:00), and evening (14:00–18:00). Error bars indicate the standard deviation of the normalized feeding intensity label. Daily values were normalized to the highest intensity detected each day, typically corresponding to the morning feeding, resulting in near zero variance for that category.

Table 1
Regression performance metrics for XGBoost and Random Forest.

Model	R^2	RMSE	MAE	95% CI for R^2
XGBoost	0.8957 ± 0.0849	0.0655 ± 0.0241	0.0425 ± 0.0184	[0.8350, 0.9564]
Random Forest	0.8961 ± 0.0835	0.0654 ± 0.0224	0.0425 ± 0.0193	[0.8363, 0.9558]

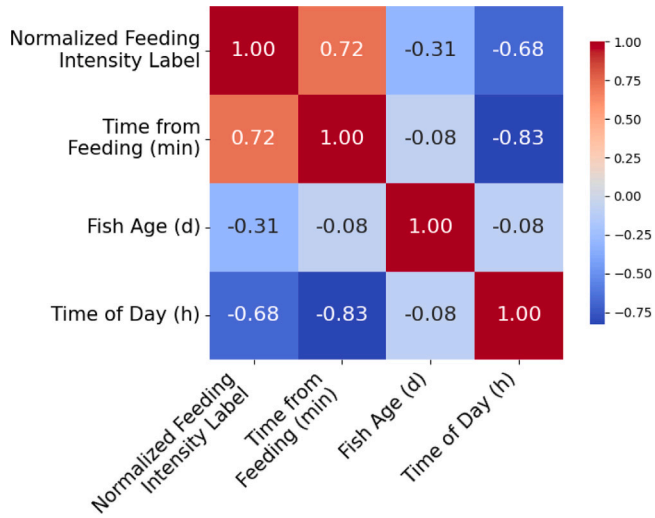

Fig. 11. Pearson correlation matrix illustrating the relationships among the numerical variables: normalized feeding intensity label, time since previous feeding (minutes), fish age (days), and time of day (hours). Positive and negative correlations indicate how feeding intensity aligns with environmental and biological factors.

A Pearson correlation analysis was conducted to examine the relationships between the environmental and biological variables, as illustrated in Fig. 11. A strong positive correlation ($r = 0.72$) was observed between normalized feeding intensity label and time from feeding, indicating that extended intervals between feedings lead to more intense feeding activity. In contrast, a negative correlation ($r = -0.68$) was found with time of day, showing that feeding intensity decreases as the day advances, a pattern consistent with typical daily activity cycles in fish. Collectively, these relationships confirm that the proposed methodology effectively captures biologically driven feeding dynamics and remains responsive to environmental variability.

Finally, regression models based on XGBoost and Random Forest were employed to examine the correspondence between the feeding intensity label and environmental and biological variables.

Both models achieved strong and consistent predictive performance, as summarized in Table 1, which presents the cross-validated R^2, RMSE, MAE, and 95% confidence intervals for each model. Reported ± values denote standard deviations obtained through 10-fold cross-validation. XGBoost and Random Forest achieved mean cross-validated R^2 values of 0.8957 ± 0.0849 and 0.8961 ± 0.0835, respectively, indicating high explainability of feeding intensity based on environmental and biological variables. Prediction errors were low, with RMSE and MAE values of 0.0655 ± 0.0241 and 0.0425 ± 0.0184 for XGBoost, and 0.0654 ± 0.0224 and 0.0425 ± 0.0193 for Random Forest. The 95% confidence intervals for R^2 were [0.8350, 0.9564] for XGBoost and [0.8363, 0.9558] for Random Forest, confirming stable and reliable model performance. Residual diagnostics further supported the robustness of the models: Shapiro–Wilk tests indicated residuals consistent with normality ($p = 0.9021$), and Breusch–Pagan tests showed no evidence of heteroscedasticity ($p = 0.9938$), demonstrating that the prediction errors exhibit no systematic deviation across the range of fitted values.

A comparison of the feature importance for normalized and unnormalized feeding intensity labels was conducted to distinguish between short-term behavioral effects and long-term growth trends, as shown in Table 2. In the normalized case, time since previous feeding emerged as the dominant predictor (importance = 0.6306), while its influence declined sharply in the unnormalized model (0.1365). In contrast, fish age gained prominence in the unnormalized label (0.4667) but contributed little under normalization (0.0925). This divergence reflects the conceptual difference between the two labels: the unnormalized version captures long-term growth trends linked to fish development, whereas the normalized label isolates short-term behavioral variation, governed primarily by the time elapsed since the previous feeding. The feature time of day maintained moderate importance across both cases (0.2767 for the normalized and 0.3967 for the unnormalized label), indicating its consistent contribution to both short-term and long-term feeding dynamics. To further verify these relationships, a permutation-based feature importance analysis was performed, yielding a consistent pattern: in the normalized label, time since previous feeding remained the dominant predictor (permutation importance = 0.7699) with fish age contributing minimally (0.1057), whereas in the unnormalized label the roles reversed, with fish age showing the strongest influence (0.9718) and time since previous feeding exhibiting only a minor effect (0.0387). Since the regression models explicitly quantify variable influence, the feature-importance results provide a clear and physically grounded decomposition of the drivers of feeding intensity. Overall, these results underscore the robustness of the proposed methodology. The feeding intensity label derived solely from acoustic data effectively encodes both environmental and biological influences, demonstrating strong agreement with expected aquaculture feeding patterns.

Table 2
Feature importance scores from the Random Forest model using normalized and unnormalized feeding intensity labels. The comparison highlights how normalization shifts model sensitivity from long-term growth trends toward short-term behavioral dynamics.

Label type	Time from feeding	Fish age	Time of day
Normalized	0.6306	0.0925	0.2767
Unnormalized	0.1365	0.4667	0.3967

4. Discussion

The method presented in this study lays the groundwork for automated feeding monitoring based on audio signals. It supports the development of advanced artificial intelligence models by leveraging optimal data reduction techniques informed by domain knowledge. This work demonstrates an advance in data reduction by transforming raw acoustic input on the order of 10^8 samples of a feeding event into a single numeric representative value of feeding intensity. This continuous regression label is sufficiently accurate to replace coarse categorical schemes, and its numerical form is central for feed optimization and welfare monitoring, since it preserves subtle within-category variation that discrete classes cannot capture.

A key advantage of this approach is its lightweight implementation: monitoring requires only a simple underwater acoustic sensor, and the matched filter consists of efficient numerical operations that can run in real time on minimal computational power, making it suitable for continuous on-site monitoring in commercial aquaculture systems. The focus of this study on a single fish species and weight group reflects commercial aquaculture practice (Bostock et al., 2010). In line with this, Symeonidou and Mente (2024) reviewed aquaculture studies and

reported that 79% employed monoculture systems, where facilities typically rear uniform cohorts to minimize competition and promote consistent growth.

Nonetheless, the low density and noncommercial experimental setting represented a relatively controlled environment and was therefore not subject to several of the complexities inherent to real aquaculture systems, such as substantial background noise generated by large water circulation systems or additional management activities, including sorting and handling. In commercial operations, tanks often contain thousands of fish at much higher stocking densities (Ellis et al., 2002), resulting in more intense feeding activity. Under such conditions, the acoustic environment becomes considerably more complex, and these factors are expected to introduce substantial background noise, thereby complicating the detection and quantification of feeding sounds. Although each tank was instrumented with two hydrophones to introduce spatial variation and capture some acoustic heterogeneity, the setup remained simpler than commercial facilities, where sensor placement, reflections, standing waves, and spatial variability in the sound field can strongly affect the reliability of feeding signal detection. The dataset itself was modest, three tanks with one to three feeding events per day over nineteen days, giving roughly 300 events across two hydrophones per tank, yet it captured meaningful variation in behavior and acoustics, and was sufficient to demonstrate the feasibility and internal consistency of the approach.

By treating the bite click template as prior biological knowledge, the machine education approach guides the matched filter to focus on the relevant signal and discount confounding sounds. The efficient propagation of sound in water, together with the underlying matched filter principle, keeps the acoustic method effective even in the presence of reflections and background activity.

With further refinement in noise reduction and appropriate adjustments, such as accounting for background noise from circulation systems and higher stocking densities in commercial tanks, the method has the potential to be effectively adapted to commercial aquaculture systems. In such settings, farmers could deploy low-cost hydrophones for real-time monitoring of feeding intensity, thereby enabling straightforward optimization of feeding routines.

5. Conclusions

This study presents a simple and robust methodology for detecting and quantifying fish feeding intensity using audio signals and domain knowledge. A matched-filter algorithm with a sliding window enables high sensitivity in detecting species-specific feeding events, using the bite sound of gilthead seabream as a reliable template. ML regression models were used to evaluate how biological and environmental parameters explain variation in the feeding label, with feature importance ranking confirming a strong explanatory value. While the effectiveness of this method in commercial fisheries has yet to be demonstrated within the scope of this article, by using only acoustic signals it meets the need for low-cost and real-time monitoring. This work serves as proof of concept, demonstrating the potential of PAM in a setup that, in many respects, resembles a commercial system. It shows that farmers could employ simple underwater audio recording equipment and minimal computing power to gain insights into feeding intensity and rapidly optimize feeding routines (Horne et al., 2024). However, real aquaculture systems introduce additional complexities, including substantial background noise from circulation equipment, the feeding activity of thousands of fish, and farm management tasks such as sorting and handling (Helberg et al., 2024); therefore, this method requires further validation under such settings.

6. Future work

This study provides a proof of concept and points toward integrating the methodology into hardware based data conversion for use in commercial farms. Future work will extend the approach to full scale operations, including hybrid models that combine matched filtering with deep feature extraction, and refine it for early disease detection and performance assessment across species, ages, environments, and feeding regimens. Subsequent research will evaluate how these developments support feed optimization, increase yield, and improve environmental sustainability.

CRediT authorship contribution statement

Dror Ettlinger-Levy: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shai Kendler:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Iris Meiri Ashkenazi:** Writing – review & editing, Methodology. **Shay Tal:** Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Barak Fishbain:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was partially funded by the Israeli Ministry of Science and Technology, grants #4348, #5543, and #6796; and the Israeli Water Authority.

Appendix A. Additional equations and data summaries

This appendix provides the formal expression for the inverse Short-Time Fourier Transform (STFT), the parameter settings used for signal preprocessing and matched filtering, and the hyperparameter grids employed in the regression analysis.

The inverse transform is given in Eq. (A.1):

$$s[t] = \mathrm{iSTFT}[f,t] = \frac{\sum_{f=0}^{L-1} \hat{S}[f,t] \cdot e^{-j2\pi f(L-t)/L} \cdot w[L-t]}{\sum_t w^2[L-t]} \tag{A.1}$$

where $s[n]$ is the input signal, $w[t]$ the analysis window centered at time t, f the frequency bin, n the time index within the window, and L is the window length.

Detailed parameter settings for the signal preprocessing and matched-filtering are summarized in Table A.1, which lists all analysis parameters applied throughout the analysis.

The normalized feeding intensity label (Fig. A.1a) shows the highest frequency of values near 1, reflecting the concentration of strong feeding events that typically occur in the morning. In contrast, noon and evening feedings are more evenly distributed, resulting in lower overall intensity and broader frequency patterns. This trend corresponds to longer intervals since the previous feeding (Fig. A.1b) and aligns with the time of day distribution (Fig. A.1c), where most feeding events cluster during morning hours. Fish age (Fig. A.1d) exhibits a wider distribution, reflecting the regular occurrence of feeding events throughout the 19 day experiment as the fish grew.

Table A.1
Core parameter settings used in signal processing, matched-filtering, and sliding window analysis.

Category	Parameter	Meaning	Value
Signal preprocessing	Window length	FFT window size	2048
	Hop length	Step size between windows	512
	Standard deviation threshold	Spectral gating sensitivity	1.5
	High-pass filter cutoff	Frequency threshold	2048 Hz
Matched-filtering and sliding window	Sliding window length	Duration of analysis window	10 min
	Sliding window step	Step size for analysis	6 s
	Detection threshold	Correlation threshold	0.01

Table A.2
Hyperparameter grid search and general settings for the regression models.

Model	Parameter	Meaning	Values searched
Random Forest	`n_estimators`	Number of trees	50, 100, 200
	`max_depth`	Maximum depth of trees	None, 5, 10, 20
	`min_samples_split`	Minimum samples to split node	2, 5, 10
XGBoost	`n_estimators`	Number of boosting rounds	50, 100, 200
	`max_depth`	Maximum depth of trees	3, 5, 10
	`learning_rate`	Step size shrinkage	0.01, 0.1, 0.2
	`subsample`	Fraction of training data sampled per tree	0.8, 1.0
General settings for both models		Train/test split	80/20
		Performance metrics	R^2 with 95% CI, RMSE, MSE

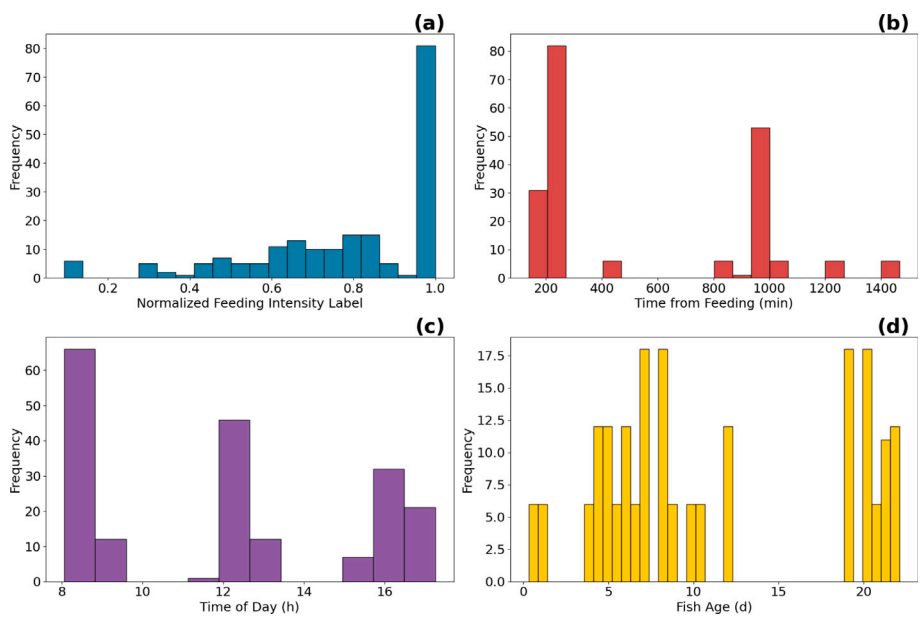

Fig. A.1. Distributions of feeding intensity peak values and environmental and biological variables across feeding events: (a) normalized feeding intensity label, (b) time since previous feeding (minutes), (c) time of day (hours), and (d) fish age in days since the start of the experiment.

Appendix B. Extraction of the gilthead seabream bite template for matched-filtering

To facilitate reproducibility, this appendix provides a detailed description of the feature extraction method used to obtain the bite template applied in the study. The template was derived from a single, manually identified seabream bite sound. The raw waveform was isolated by examining feeding activity 5–10 min after the start of the feeding event, when individual bites could be more easily distinguished due to the natural decay in biting frequency after most of the food had been consumed. The template was validated against video recordings and by auditory inspection. Once the raw waveform was obtained, the template was preprocessed in the same way as the signal, using spectral gating and high-pass filtering for noise reduction (see Table A.1 for parameter settings). The final template was defined as the waveform segment with a sharp impulsive onset, corresponding to the clicking sound produced when a pellet was bitten, as shown in Fig. 3.

Appendix C. Sensitivity analysis of the matched-filter to overlapping bite signals

To evaluate the sensitivity of the matched-filter to overlapping bite signals, artificial waveforms were generated by combining two to eight seabream bite templates with delays ranging from 1 to 15 ms. Fig. C.1 shows an example of three overlapping templates separated by 5 ms, while Fig. C.2 presents the corresponding heat map of MF responses. In the heat map shown in the x-axis denotes the delay between templates, the y-axis indicates the number of overlapping templates, and the color scale represents the ratio between the MF response of overlapping and single template signals, where the single-signal response originates from an individual, non-overlapping bite event.

Throughout the tested range, the MF maintained a response ratio above 0.8, confirming its robustness under moderate overlap. A slight increase observed near a 2 ms delay likely occurs because overlapping signals add together when their waves align, making the overall

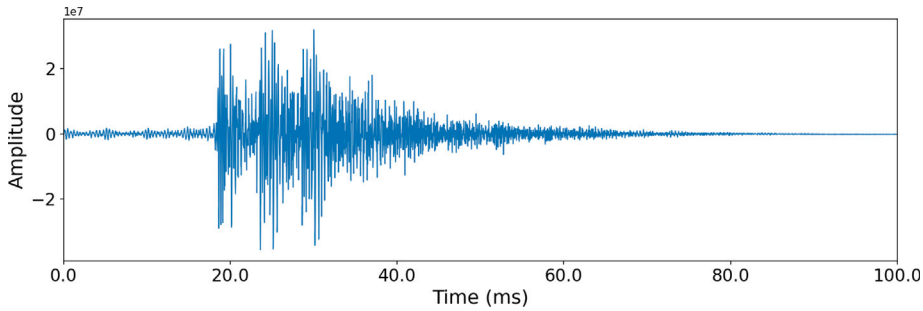

Fig. C.1. Example of an artificially generated overlapping bite signal composed of three seabream bite templates spaced 5 ms apart. The waveform illustrates how closely timed feeding events combine to produce a complex signal pattern, resembling conditions in which multiple fish feed simultaneously in aquaculture environments.

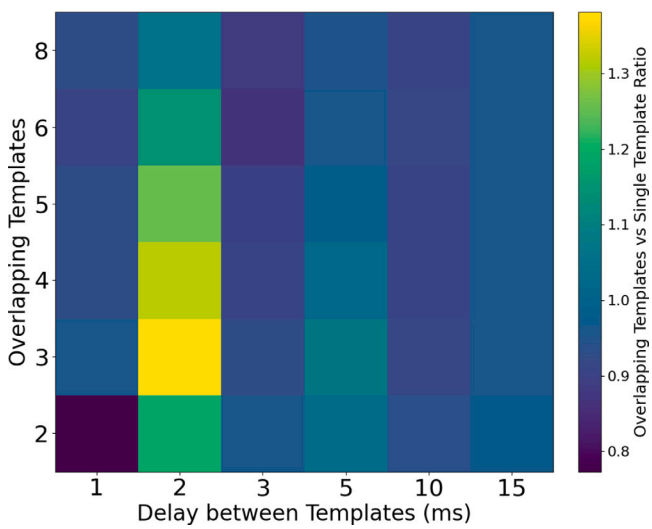

Fig. C.2. Heatmap showing the effect of overlapping bite templates on matched-filter response. The x-axis represents the delay between templates, the y-axis shows the number of overlapping templates, and the color scale indicates the ratio between the matched-filter output for overlapping templates and a single template. Ratios above 0.8 across most conditions demonstrate that the matched-filter maintains reliable detection even under moderate overlap levels.

response slightly stronger. Overall, the effect of overlap reduced MF accuracy by no more than 20%, demonstrating that the method remains reliable for quantifying feeding intensity in realistic, noisy environments. For aquaculture systems with high fish density, it is possible to lower the method's sensitivity to gain higher temporal resolution. In addition, the feeding intensity can also be defined using the sum of matched-filter values exceeding the threshold instead of counting the number of samples above it. In this study, however, the method was adjusted to the conditions of small experimental tanks operating in non-industrial settings. Ultimately, because detections rely on MF responses surpassing a defined threshold, overlapping signals yielding 80% of the reference amplitude are still correctly identified, maintaining full detection accuracy.

Data availability

Data will be made available on request.

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